

**YEAR 8 SCIENCE
CHEMICAL SCIENCES
REVISION - TEST 1**

1. Look at p. 230 Science Quest 8 and use it to answer the following about the development of our current model of the atom. For each person, list what they found about the atom.

Democritus (500 BC): everything is made up of atoms

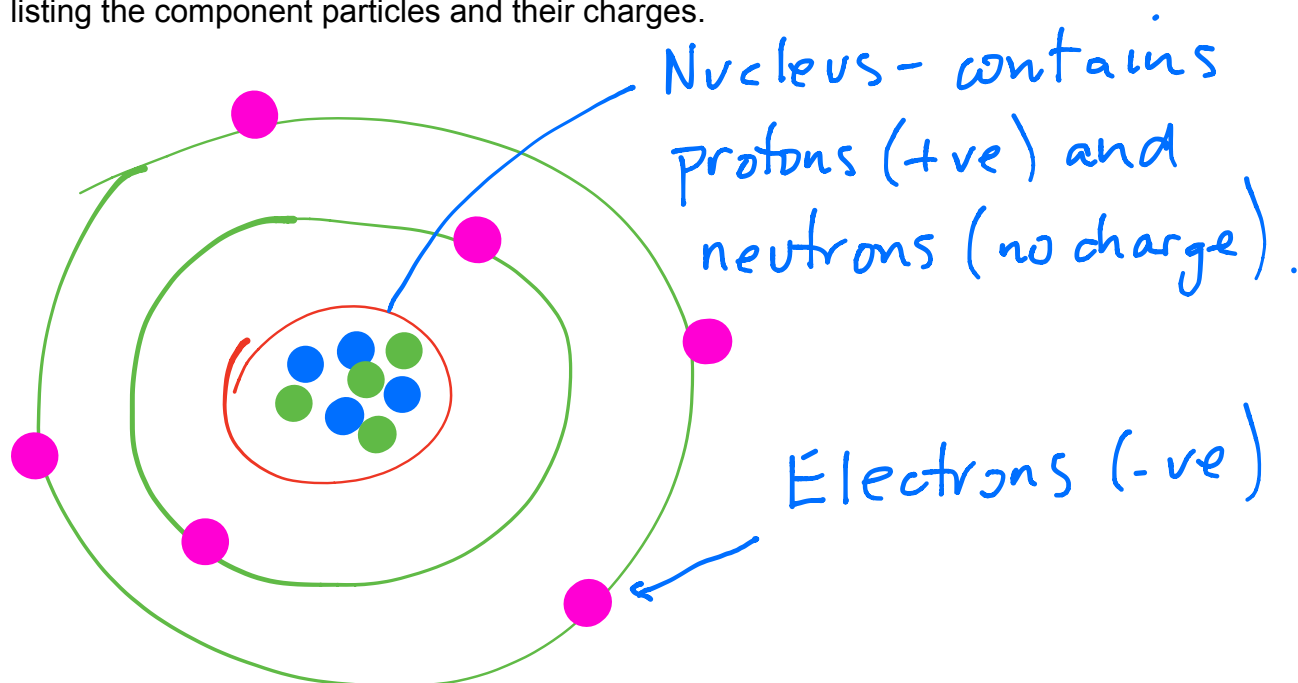
Dalton (1803): matter is made up of atoms.

Rutherford (1911): proved atoms are made up of lots of space.

Bohr (1931): electrons move around the nucleus.

Chadwick (1932): found neutrons in the nucleus.

2. Describe the structure of an atom. Draw a labelled diagram to help your explanation, listing the component particles and their charges.



3. Define the following terms.

Element: contains atoms that are all the same

Compound: made up of 2 or more elements.

Mixture: 2 or more substances together.

Atomic number: number of protons.

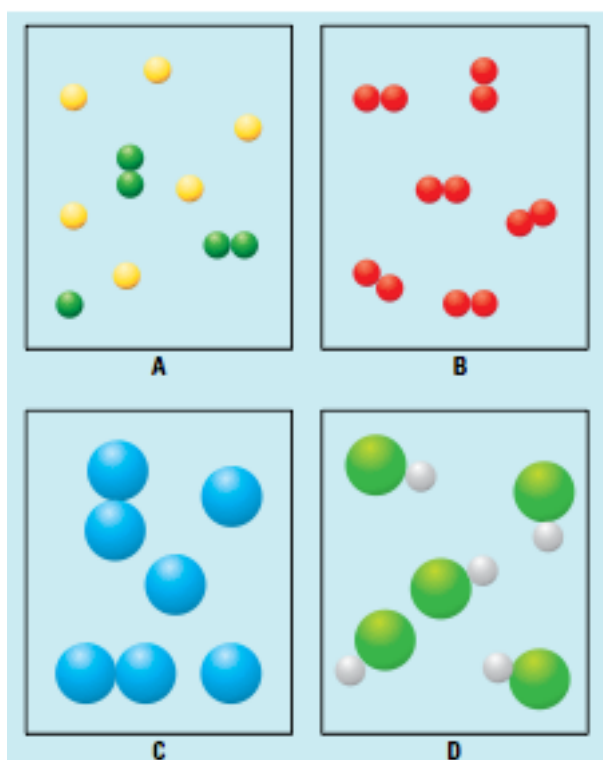
Atomic mass: number of protons + neutrons.

4. Which of the diagrams below represent:

(a) elements. B + C

(b) compounds. D

(c) mixtures. A



5. Classify the following substances as element, compound or mixture.

~~H₂O~~ ~~air~~ ~~H₂~~ ~~SiO₂~~ cup of tea ~~gold~~ ~~mercury oxide~~

Element	Compound	Mixture
H ₂ gold	H ₂ O SiO ₂ mercury oxide	tea air (N ₂ + O ₂)

6. Complete the following table, either with the name or symbol of the element.

ELEMENT	SYMBOL
aluminium	Al
Sodium	Na
nitrogen	N
boron	B
zinc	Zn
lead	Pb
iron	Fe
chlorine	Cl
potassium	K
neon	Ne

ELEMENT	SYMBOL
tin	Sn
copper	Cu
silicon	Si
fluorine	F
helium	He
magnesium	Mg
silver	Ag
gold	Au
mercury	Hg
bromine	

7. Name the elements found in the following compounds and give the number of atoms of each.

COMPOUND	NAME AND NUMBER OF ELEMENTS PRESENT
CH ₃ COOH	2 × C, 4 × H, 2 × O
Al ₂ O ₃	2 × Al, 3 × O
H ₂ SO ₄	2 × H, 1 × S, 4 × O

8. Complete the following table from the information presented about the Periodic Table.

	Group 1	Group 2	
Period 1	1.0 H Hydrogen 1		Relative atomic mass Symbol Atomic number
Period 2	6.9 Li Lithium 3	9.0 Be Beryllium 4	
Period 3	23.0 Na Sodium 11	24.3 Mg Magnesium 12	

$P + n = A$

Element	Symbol	Atomic number	Atomic mass	Number of protons	Number of electrons	Number of neutrons
Hydrogen	H	1	1	1	1	0
Lithium	Li	3	7	3	3	4
Beryllium	Be	4	9	4	4	5
Sodium	Na	11	23	11	11	12
Magnesium	Mg	12	24	12	12	12